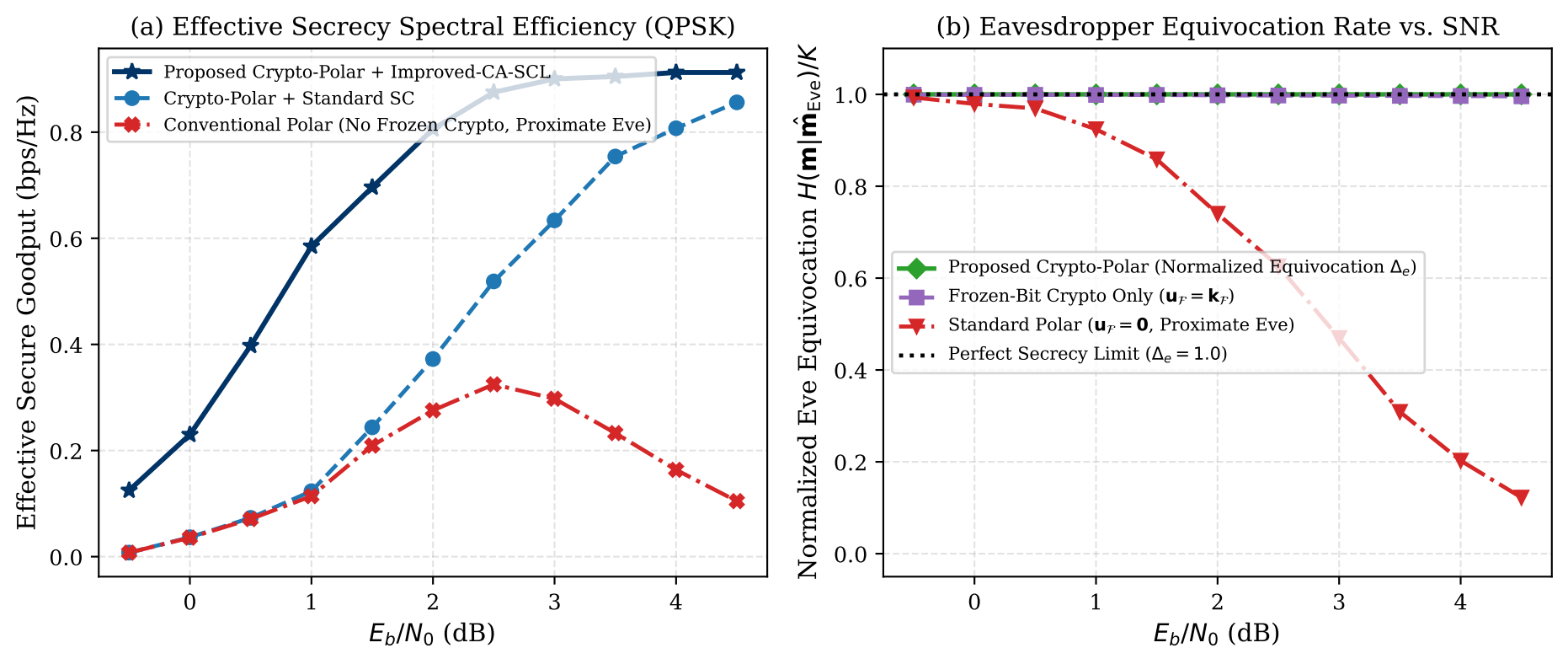
(a) Effective Secrecy Spectral Efficiency (QPSK)
Proposed Crypto-Polar + Improved-CA-SCL
Crypto-Polar + Standard SC
Conventional Polar (No Frozen Crypto, Proximate Eve)
Effective Secure Goodput (bps/Hz)
E_b/N_0 (dB)
(b) Eavesdropper Equivocation Rate vs. SNR
Proposed Crypto-Polar (Normalized Equivocation Δ_e)
Frozen-Bit Crypto Only ($\mathbf{u}_{\mathcal{F}} = \mathbf{k}_{\mathcal{F}}$)
Standard Polar ($\mathbf{u}_{\mathcal{F}} = \mathbf{0}$, Proximate Eve)
Perfect Secrecy Limit ($\Delta_e = 1.0$)
Normalized Eve Equivocation $H(\mathbf{m}|\hat{\mathbf{m}}_{\text{Eve}})/K$
E_b/N_0 (dB)